

Olist Brazilian E-Commerce

Marketplace Performance & Customer Satisfaction Analysis

1 Business Background & Problem Statement

Olist operates as a **marketplace integrator** in Brazil: independent small and medium merchants sign a single contract with Olist in order to sell across multiple large online marketplaces without negotiating each sales channel individually. When a customer places an order through an Olist-connected storefront, the responsible seller ships the item through Olist's logistics partners. After the order is delivered — or after the estimated delivery date has passed — the customer automatically receives a satisfaction survey and can leave a 1-to-5 star review, optionally with a written comment.

1.1 The Business Challenge

Olist's leadership has approximately two years of order history spanning orders, order items, payments, products, sellers, customers, reviews, and geolocation records — but **no single consolidated analysis** currently connects these dimensions. Review scores, delivery timing, seller performance, and payment behavior all vary simultaneously across time, product category, and Brazilian state, making it difficult to separate genuine root causes from coincidental correlations. Leadership needs a clear, evidence-based picture of three things:

- **How the marketplace is actually performing** — is growth in orders and revenue translating into a healthier platform, or is it masking operational strain?
- **What is shaping customer satisfaction and dissatisfaction** — which factors are primary drivers of review scores, and which are secondary or coincidental?
- **Where the clearest opportunities are to improve the platform as Olist scales** — which levers (logistics, seller management, geography, category strategy) offer the highest-leverage fixes?

1.2 Objective of the Analysis

The analysis is structured to move through **Data Understanding** → **Cleaning** → **Feature Engineering** → **Exploratory Analysis** → **Six Core Business Questions** → **a supporting Machine-Learning model** → **Business Recommendations**, fully reproducible end-to-end. This is real operational data rather than a synthetic dataset built around a predetermined story — there is no target/label column and no single “correct” answer. The goal is a well-reasoned, evidence-based narrative that leadership can act on, not a forced conclusion.

1.3 The Six Core Questions

#	Core Question	What It Resolves
1	Marketplace Performance Over Time	How have order volume, revenue, and review scores trended across Sep 2016–Oct 2018? Do they move together?
2	Delivery Performance & Satisfaction	How does delivery timing relative to the estimated date relate to review scores, and does this hold across categories/regions?
3	Seller & Geographic Patterns	How does the geographic distribution of sellers and customers relate to delivery, freight cost, and satisfaction?
4	Product Category Performance	How do order volume, price, and review scores differ across product categories?
5	Payment Behavior	How do payment type and installment choices relate to order value and to customer experience?
6	Root Cause Analysis	What factors most strongly associate with low review scores — and which are primary vs. secondary?

In addition to the six core questions, four supporting deep-dives are scoped: repeat-customer / retention behavior, freight cost vs. product characteristics, seller-level performance variance, and anomaly detection.

2 Source Datasets & Data-Quality Assessment

The Brazilian E-Commerce Public Dataset by Olist (Kaggle, CC BY-NC-SA) is supplied as **nine relational tables**, joined primarily through `order_id`, `customer_id`, `product_id`, and `seller_id`. The relationship structure is:

customers → orders → order_items → products → product_category_name_translation
order_items → sellers
orders → order_payments
orders → order_reviews
customers / sellers → geolocation (matched by zip-code prefix — <i>not</i> a strict foreign key)

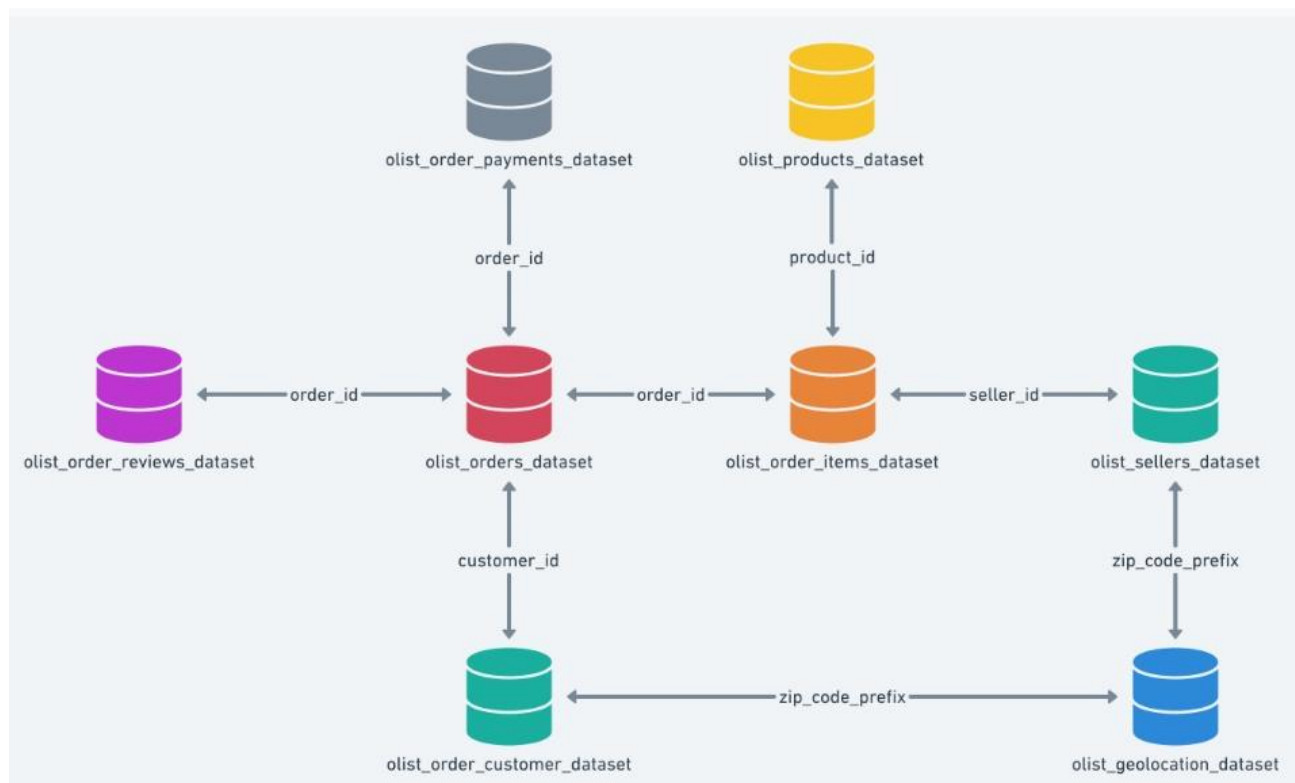
2.1 Table Inventory

Table	Grain	Key Fields	Role in Analysis
orders	1 row / order	order_id, customer_id, order_status, 4 lifecycle timestamps, estimated delivery date	Backbone of the order-grain master table; source of delivery timing features
order_items	1 row / item in an order	order_id, product_id, seller_id, price, freight_value	Aggregated to order grain (sum price/freight, count items) before joining
order_payments	1+ rows / order	payment_type, installments, payment_value	Aggregated to order grain; drives payment-behavior analysis
order_reviews	1 row / review	review_score, comment_title/message, survey timestamps	Target variable for satisfaction and root-cause analysis
products	1 row / product	category name, weight, dimensions	Category performance and freight/product-characteristic analysis
sellers	1 row / seller	seller_id, zip prefix, city, state	Seller geography and seller-level performance variance
customers	1 row / order (see 2.3)	customer_id, customer_unique_id, zip prefix, city, state	Customer geography and repeat-purchase / retention analysis
geolocation	~53 rows / zip prefix	zip prefix, lat/lng, city, state	Aggregated to 1 avg. (lat, lng) per prefix; distance-proxy feature
category_name_translation	1 row / category	Portuguese → English category name	Lookup table for readable category labels

Overall scope: **99,441 orders** across the nine tables, covering **September 2016 – October 2018**.

Dataset Relationship Schema

Fig. 2.1 — Entity-relationship diagram across the nine Olist datasets, keyed primarily by `order_id`, `customer_id`, `product_id`, and `seller_id` (`zip_code_prefix` joins are non-strict, matched approximations).



Arrows indicate join direction; each dataset icon corresponds to one of the nine source tables described in Section 2.1.

2.2 Data-Quality Issues Identified

A missingness and consistency scan surfaces the following characteristics, each of which drives a specific cleaning decision in Section 3:

Issue	Detail	Downstream Handling
Incomplete order lifecycle	order_approved_at, order_delivered_carrier_date, and order_delivered_customer_date all contain missing values, consistent with orders that never completed the normal fulfillment flow (canceled, unavailable, still processing).	Non-delivered orders are kept for volume/revenue/status analysis but excluded from delivery-timing and review-driver analysis.
Missing product listing data	~610 products are missing product_category_name and related listing fields; a small number are missing weight/dimension fields.	Missing category is labeled "unknown" rather than dropped, preserving ~610 products' order volume.
Sparse free-text reviews	review_comment_title and review_comment_message are missing for the large majority of reviews — most customers rate without writing anything.	Free-text analysis is scoped as optional/secondary; review_score (always present) is the primary target.
Dual customer identifiers	customer_id is one row per order (99,441 unique); customer_unique_id identifies the real person (96,096 unique) — implying 3,345 orders come from repeat shoppers.	customer_unique_id is used for any repeat-purchase/retention analysis; customer_id is used only as the order-level join key.
Multi-item / multi-payment orders	9,803 orders contain more than one order_items row; 2,961 orders contain more than one payment record.	Both are aggregated (summed) to the order grain before joining, so order counts are never inflated.
Geolocation duplication	1,000,163 geolocation rows span only 19,015 unique zip prefixes — an average of ~53 readings per prefix, which would multiply rows on a naive join.	Geolocation is aggregated to one averaged (lat, lng) per zip-code prefix before joining to customers/sellers.
Non-strict geo join	Zip-code prefixes are matched to geolocation rather than a strict foreign key, so coordinates are an approximation, not an exact address.	Distances derived from this join are treated as a proxy, not a precise measure, in all interpretation.

2.3 Fulfillment Funnel Snapshot

96,478 of 99,441 orders (97.0%) carry status 'delivered'; the remaining ~3% are canceled, unavailable, in transit, or otherwise unresolved as of the data extract. This 97% 'delivered' base is the population used for all delivery-timing and review-driver analysis described in Section 3.

3 Proposed Data-Analysis Approach — Overview

The proposed approach follows a single, reproducible pipeline: every downstream question is answered from **one order-grain master table**, built once and reused throughout, rather than from ad-hoc joins repeated per question. The pipeline has three layers, expanded stage-by-stage in Sections 4–14:

Layer	Stages	Purpose
A — Foundation	Data Cleaning → Data Modeling → KPI Framework	Produce one trustworthy, feature-rich order-grain table and define the metrics every later stage reports against.
B — Dimensional Analysis	Sales → Delivery → Customer Satisfaction → Retention → Product/Category → Seller → Geography → Payments	Answer the six core questions plus supporting deep-dives, one business dimension at a time, cross-referencing findings across dimensions.
C — Synthesis	Statistical / Root-Cause Analysis → Predictive Model → Recommendations	Separate primary drivers from secondary/contributing factors and translate findings into a specific, actionable business intervention.

3.1 Guiding Principles

- **One master table, many views** — every analysis stage below reads from the same cleaned, order-grain table so figures are consistent across sections.
- **Distinguish primary vs. secondary drivers** — every finding is tested for whether it holds independently or only because it correlates with something else (e.g., category effects that are really delivery-difficulty effects).
- **Evidence over narrative** — this is real operational data with no predetermined story; conclusions follow from the data, and ambiguous results are reported as ambiguous.
- **Actionability** — each stage closes with a specific business takeaway, and the final model is scoped to information available *before* the outcome is known, so it can drive real intervention rather than hindsight explanation.

4 Stage 1 — Data Cleaning

STEP 1 | Data Cleaning & Preparation

Objective: Resolve the data-quality issues in Section 2.2 with explicit, documented rules before any joining or analysis begins.

- Parse all timestamp columns to proper datetimes at load time (order purchase/approval/carrier/delivery/estimate, review creation/answer).
- Retain non-delivered orders for volume/revenue/status reporting, but exclude them from delivery-timing and review-driver analysis (they lack complete delivery timestamps by construction).
- Aggregate order_items to the order grain (sum price, sum freight, count items) and order_payments to the order grain (sum payment_value, max installments, primary payment_type) before any join, so order counts are never inflated by multi-item or multi-payment orders.
- Aggregate the ~53-reading-per-prefix geolocation table to one averaged (latitude, longitude) per zip-code prefix before joining to customers/sellers.
- Translate product_category_name to English via the category-translation lookup; label missing categories “unknown” rather than dropping ~610 products’ order volume.
- Standardize customer_unique_id as the sole key for any unique-shopper / repeat-purchase logic; reserve customer_id strictly for order-level joins.
- Flag and separately review outlier orders (extreme delivery delay, extreme price) rather than silently clipping them, since they carry diagnostic value (Section 11.4).

Output: A validated set of cleaning rules with before/after row counts and a written data-quality log, ready to feed the modeling stage.

5 Stage 2 — Data Modeling & Feature Engineering

STEP 2 | Building the Order-Grain Master Table

Objective: Join all nine cleaned tables into a single order-grain analytical table and engineer the features every downstream question needs.

- Join sequence: customers → orders → (aggregated) order_items → products → category translation → sellers → (aggregated) order_payments → order_reviews → averaged geolocation (matched twice, once for customer zip and once for seller zip).
- Engineer delivery_delay_days (actual vs. estimated delivery date, negative = early) and delivery_lead_days (purchase to delivery, total transit time).
- Engineer seller_handling_days (purchase to carrier hand-off) as the controllable portion of lead time.
- Engineer order_month/order_year for time-series aggregation; price-per-item and freight-per-item for basket-size-normalized comparisons.
- Engineer a customer–seller distance proxy from the averaged lat/long pair (degrees, not km — a directional proxy given the non-strict geo join).
- Engineer review-score buckets and a delivery-timing bucket (7+ days early ... 7+ days late) used throughout Section 8.
- Document the master table's grain explicitly (one row per order_id) and its row count against the source orders table, so no downstream aggregation silently double-counts.

Output: One documented, order-grain master table (~99,441 rows) with all engineered features, used as the single source for every stage from here forward.

6 Stage 3 — KPI Framework

Before running dimensional analysis, a fixed set of KPIs is defined so every later section reports against the same baseline figures rather than re-deriving ad-hoc metrics. KPIs are grouped by business area:

Marketplace / Sales

Monthly order volume & revenue (order price + freight); month-over-month growth rate; average order value; seasonality (e.g., November peak).

Delivery

% orders delivered late vs. early; average delivery delay (days); average total lead time; average seller handling time.

Customer Satisfaction

Platform-average review score; review-score distribution (1–5 share); average review score by delivery-timing bucket.

Retention

Repeat-purchase rate (customer_unique_id basis); average lifetime value, one-time vs. repeat buyers; orders per unique customer.

Product / Category

Order volume and revenue share by category; average review score by category; average price by category.

Seller

Orders per seller; seller on-time rate; seller average review score (min. volume threshold applied); seller count and concentration by state.

Geography

Orders and sellers by state; average delivery delay by state; average freight cost by state; same-state vs. cross-state performance gap.

Payments

Payment-type mix (share of orders); average installment count; average order value by payment type; average review score by payment type.

7 Sales & Marketplace Performance Analysis

Guiding question: How have order volume, revenue, and review scores trended across the available time period, and do they tell the same story?

Analytical approach

- Aggregate the master table to monthly order volume and monthly revenue (order price + freight); plot as a time series alongside average monthly review score on a shared timeline.
- Identify seasonal spikes (e.g., a Black-Friday-adjacent November peak) and the launch ramp-up period at the start of the dataset.
- Check whether review-score trend moves independently of the volume/revenue trend, or whether periods of rising volume coincide with softening reviews.

Supporting visuals

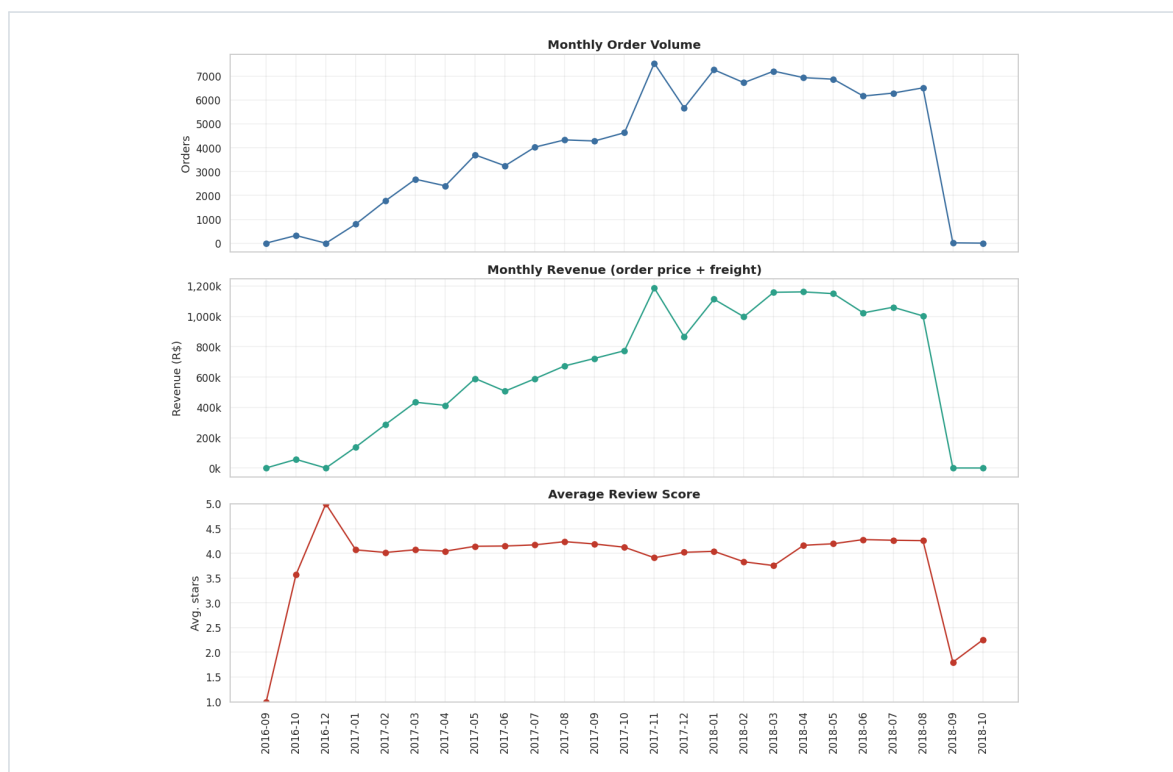


Fig. 7.1 — Monthly order volume, revenue, and average review score, Sep 2016 – Oct 2018. The review-score line moves largely independently of the volume/revenue trend.

What the output is expected to reveal: Volume and revenue are expected to show near-continuous growth as Olist scales onto more channels, while the review-score line should be tracked as a separate KPI rather than assumed to move with growth — a flat-or-declining review trend during rising volume signals an operations bottleneck rather than a demand problem.

8 Delivery Performance Analysis

Guiding question: How does delivery timing relative to the estimated delivery date relate to review scores, and does the relationship hold across categories and regions?

Analytical approach

- Bucket every delivered order by delivery-timing (7+ days early ... 7+ days late) and compute average review score and order count per bucket against the platform-average line.
- Cross-tabulate delivery delay against review score directly (scatter/jitter view) and compute the share of late deliveries within each review-score tier (1–5).
- Build category×delivery-timing and state×delivery-timing heatmaps of average review score to test whether the delay–satisfaction relationship is universal or concentrated in specific categories/regions.

Supporting visuals

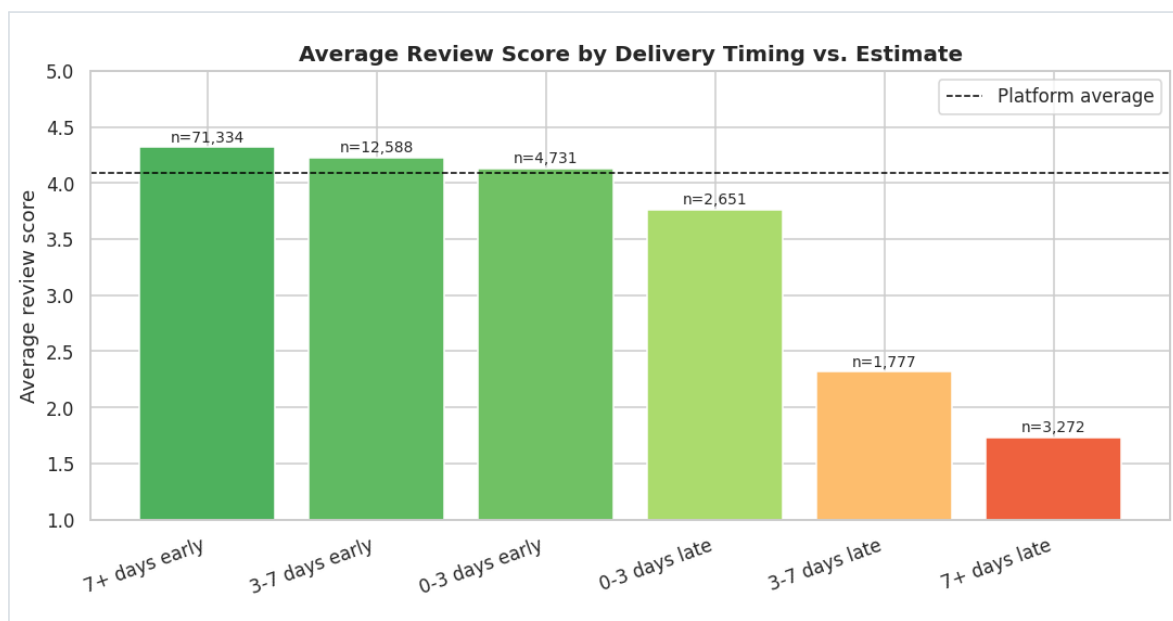


Fig. 8.1 — Average review score by delivery-timing bucket vs. the platform average. Score drops sharply once a delivery is late.

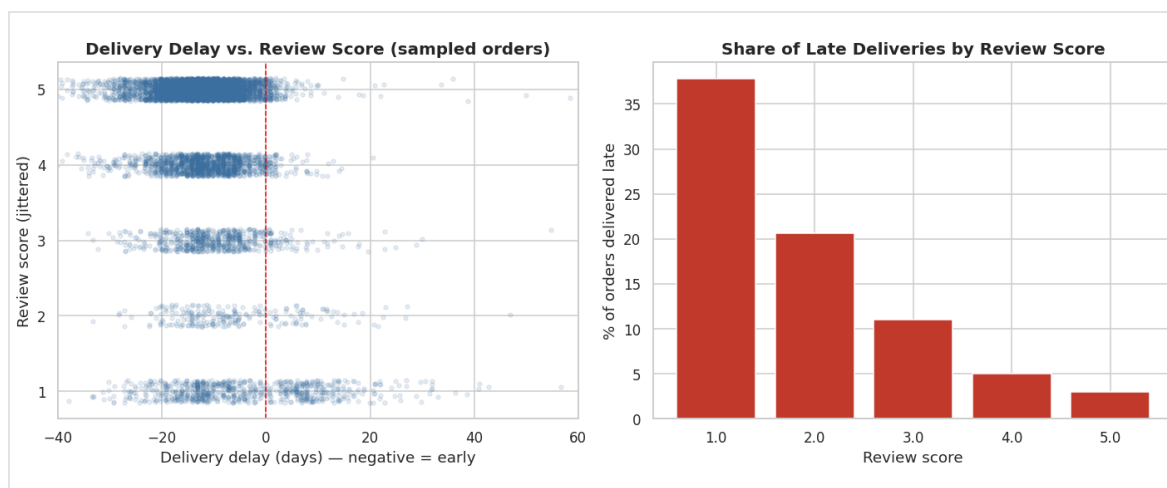


Fig. 8.2 — Delivery delay vs. review score, jittered (left), and share of orders delivered late by review score (right).

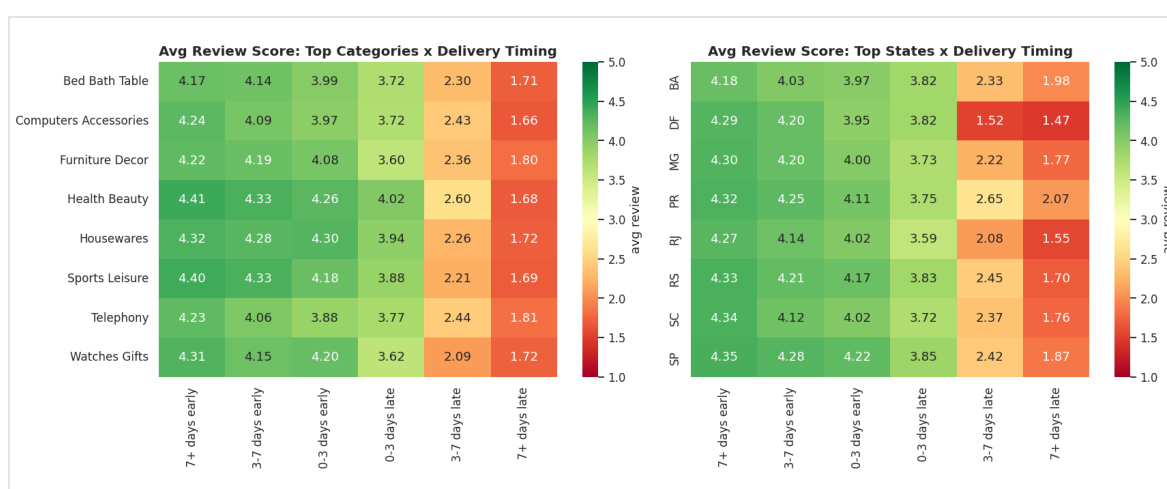


Fig. 8.3 — Average review score across delivery-timing buckets, broken out by top product categories (left) and top states (right).

What the output is expected to reveal: Average review score is expected to fall sharply and near-monotonically as deliveries move from early to late, with the effect holding broadly across nearly all categories and states — identifying this as a systemic logistics issue rather than a localized one, and the single strongest bivariate relationship in the dataset.

9 Customer Satisfaction Analysis

Guiding question: *Beyond delivery timing alone, what does the overall shape of customer satisfaction look like, and how consistent is it across the platform?*

Analytical approach

- Establish the platform-wide review-score distribution and average as the baseline against which every other cut (category, state, seller, payment type) is compared.
- Layer in the delivery-timing findings from Section 8 as the primary satisfaction lens, then test category, geography, seller, and payment cuts as secondary lenses.
- Separate written-review analysis (a minority of orders, given sparse free-text) as an optional qualitative supplement, not a primary metric.

What the output is expected to reveal: *Satisfaction should emerge as a composite outcome dominated by delivery experience, with category, geography, and payment behavior contributing smaller, largely dependent effects rather than independent drivers.*

10 Retention & Repeat-Customer Analysis

Guiding question: What does the dataset reveal about repeat-purchase behavior, and how valuable are repeat customers relative to one-time buyers?

Analytical approach

- Use `customer_unique_id` (not `customer_id`) to correctly count unique shoppers and compute the true repeat-purchase rate over the ~2-year window.
- Compare average lifetime spend for one-time buyers vs. repeat customers.
- Cross-reference repeat-purchase rate against the delivery/satisfaction findings from Sections 8–9 to assess whether poor delivery experience plausibly suppresses repeat behavior.

Supporting visuals

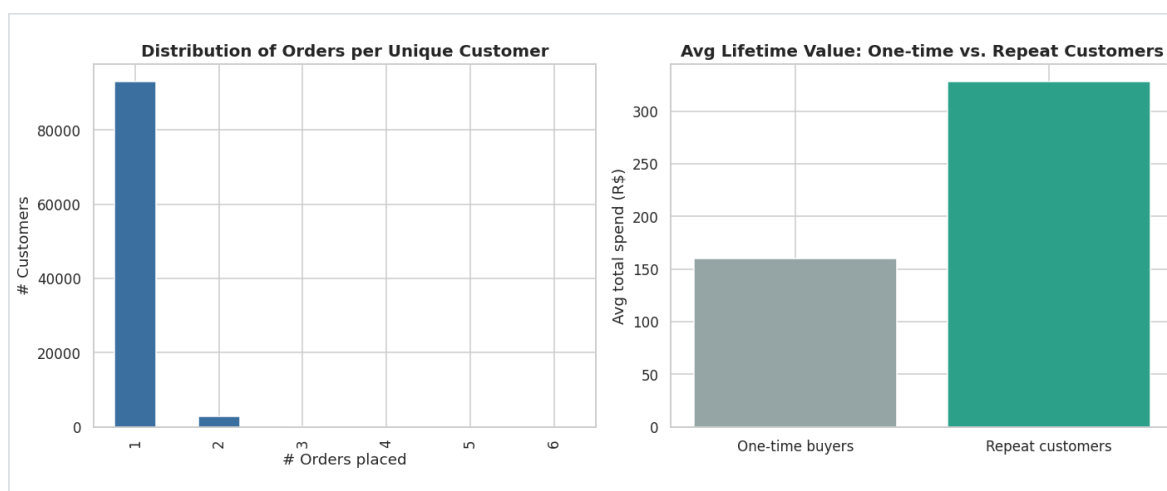


Fig. 10.1 — Distribution of orders placed per unique customer (left) and average lifetime spend, one-time vs. repeat customers (right).

What the output is expected to reveal: Repeat-purchase rate is expected to be very low (well under 5% of unique shoppers), consistent with a high-churn, acquisition-driven marketplace rather than a loyalty-driven one — positioning delivery-reliability improvements as a plausible retention lever, not just a satisfaction metric.

11 Product & Category Performance Analysis

Guiding question: How do order volume, price, and review scores differ across product categories, and are any categories notably stronger or weaker than the platform average?

Analytical approach

- Rank categories by order volume and revenue share to characterize the long-tail distribution of the catalog.
- Compute average review score per category and compare against categories' late-delivery rates to test whether low-review categories are simply harder to ship (bulky/heavy items).
- Build a price-vs-satisfaction bubble view (bubble size = order volume) to test whether price level relates to satisfaction independent of delivery.
- Flag statistical anomalies: categories or individual orders with extreme delay or pricing that warrant separate investigation rather than blending into category averages.

Supporting visuals

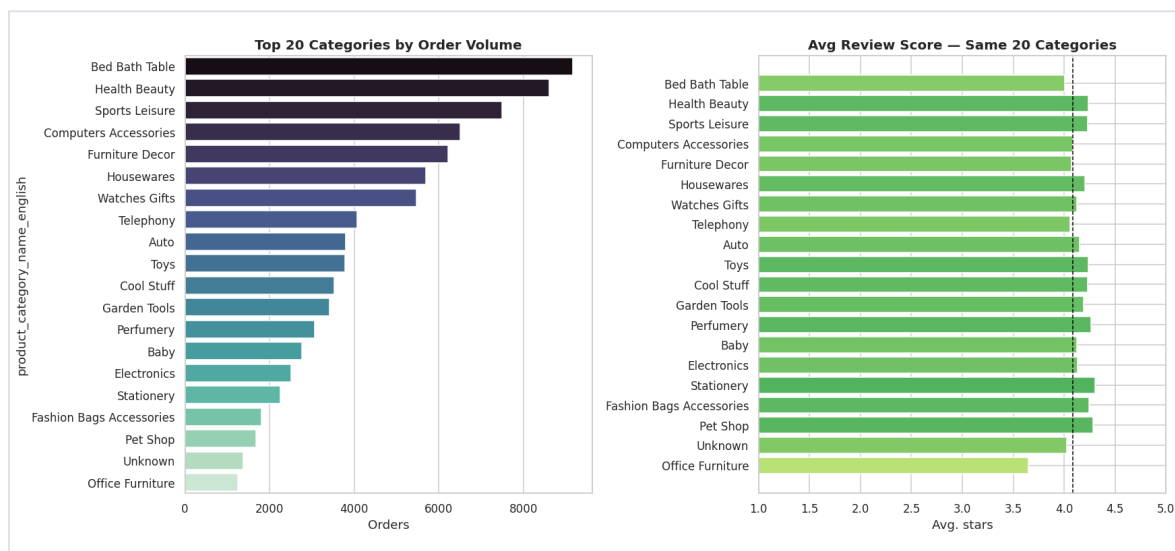


Fig. 11.1 — Top 20 categories by order volume (left) and their average review scores (right).

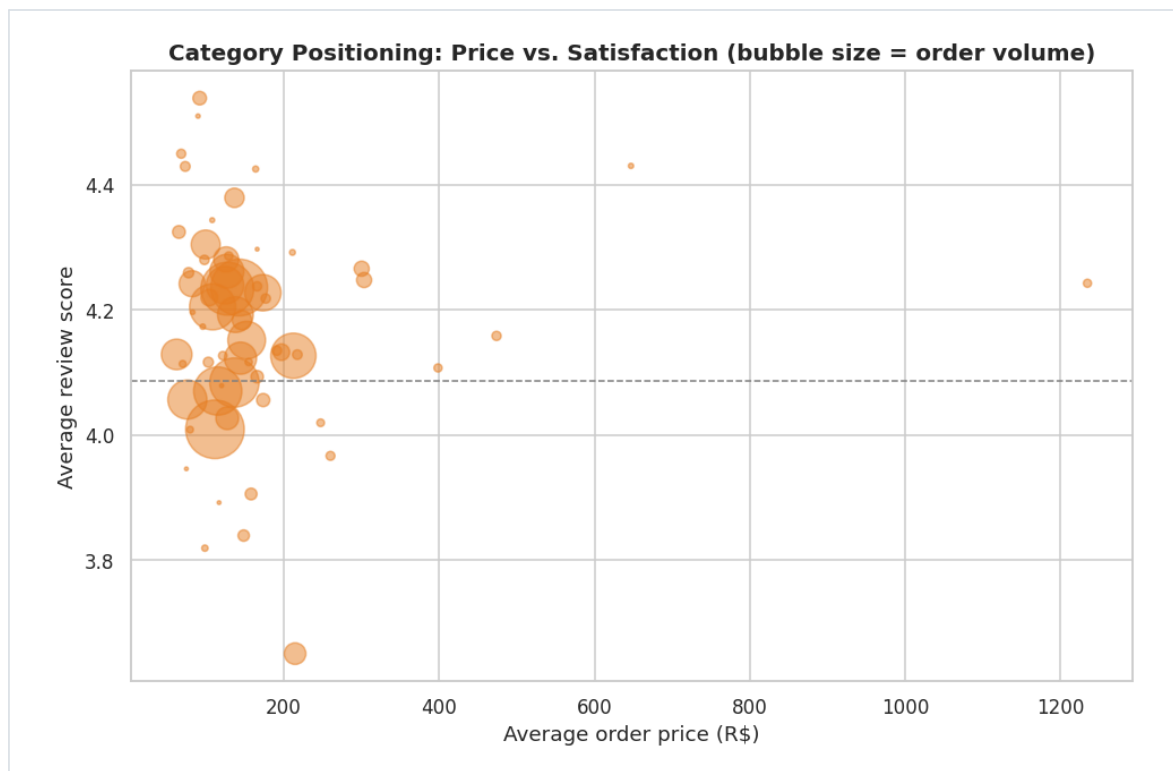


Fig. 11.2 — Category positioning: average order price vs. average review score, bubble size = order volume.

What the output is expected to reveal: Category-level review differences are expected to track closely with each category's shipping difficulty (weight/bulk) rather than price or demand — making category performance a downstream symptom of delivery risk more than an independent driver.

11.1 Supporting Deep-Dive: Freight vs. Product Characteristics

Since bulky/heavy categories surface repeatedly as lower-scoring, freight cost is tested directly against product weight and volume (rather than distance alone) to confirm whether shipping difficulty — not just distance — is the mechanism behind category-level dissatisfaction.

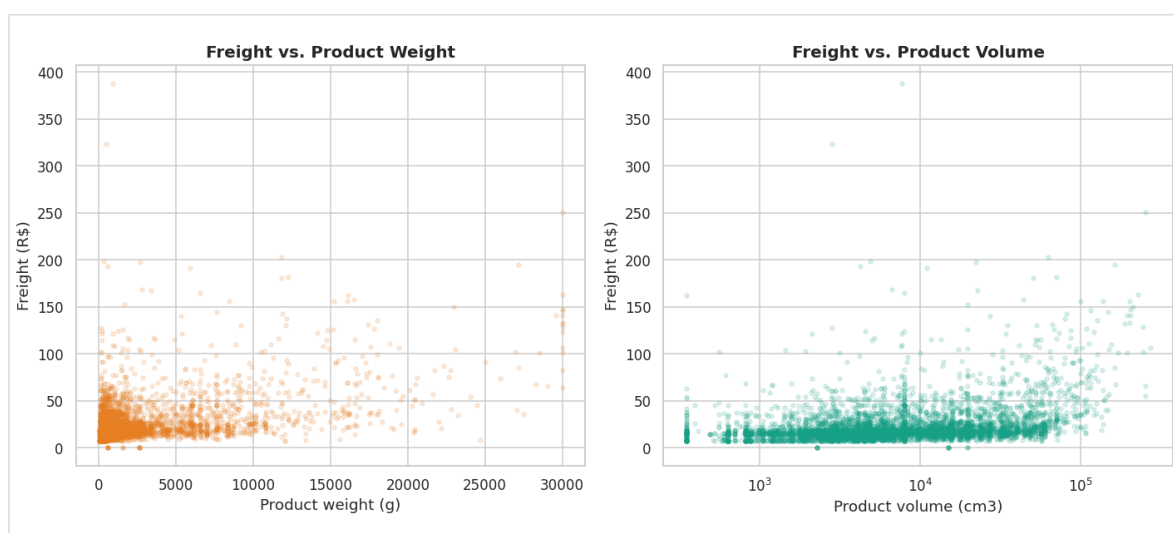


Fig. 11.3 — Freight cost vs. product weight (left) and vs. product volume, log scale (right).

Freight cost is expected to correlate more strongly with weight/volume than with raw customer–seller distance — confirming that bulky-category dissatisfaction is intertwined with genuinely higher shipping cost and complexity, not just longer shipping

distances.

12 Seller Performance & Concentration Analysis

Guiding question: Beyond category and geography, do individual sellers show meaningfully different performance, and how concentrated is the seller base?

Analytical approach

- Compute orders-by-seller-state and compare seller concentration against customer/order concentration by state to quantify geographic imbalance.
- Build a seller scorecard (minimum order-volume threshold applied) combining on-time delivery rate and average review score per seller.
- Plot the distribution of average review score across qualifying sellers, and seller late-delivery rate vs. average review (bubble = order volume) to locate consistently underperforming sellers.

Supporting visuals



Fig. 12.1 — Distribution of average review score across sellers with ≥ 15 delivered orders (left) and seller late-delivery rate vs. average review, bubble size = order volume (right).

What the output is expected to reveal: A real, seller-specific performance gap is expected beyond what category and geography alone explain, with higher late-delivery-rate sellers clustering at the low end of the review distribution — supporting a seller scorecard as an operational tool for coaching or deprioritization.

13 Geography Analysis

Guiding question: How does the geographic distribution of sellers and customers across Brazilian states relate to delivery performance, freight cost, and satisfaction?

Analytical approach

- Compare orders-by-customer-state against sellers-by-state to quantify how much more concentrated the seller base is than the customer base.
- Compute average delivery delay, average freight cost, and average review score by state (top states by volume) side by side.
- Test freight cost against the customer–seller distance proxy, and compare same-state vs. cross-state shipments on delay, freight, and review score.

Supporting visuals

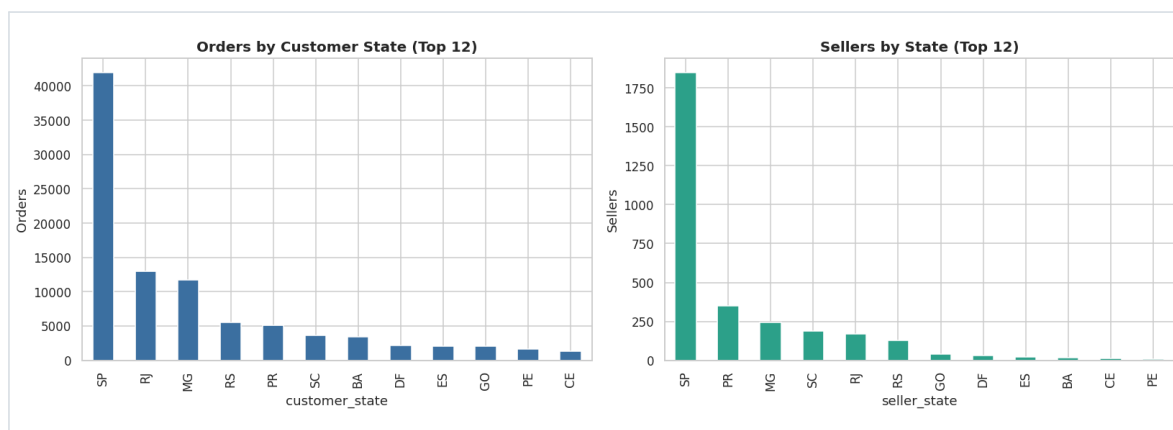


Fig. 13.1 — Orders by customer state (left) vs. sellers by state (right), top 12 by volume. The seller base is far more geographically concentrated.

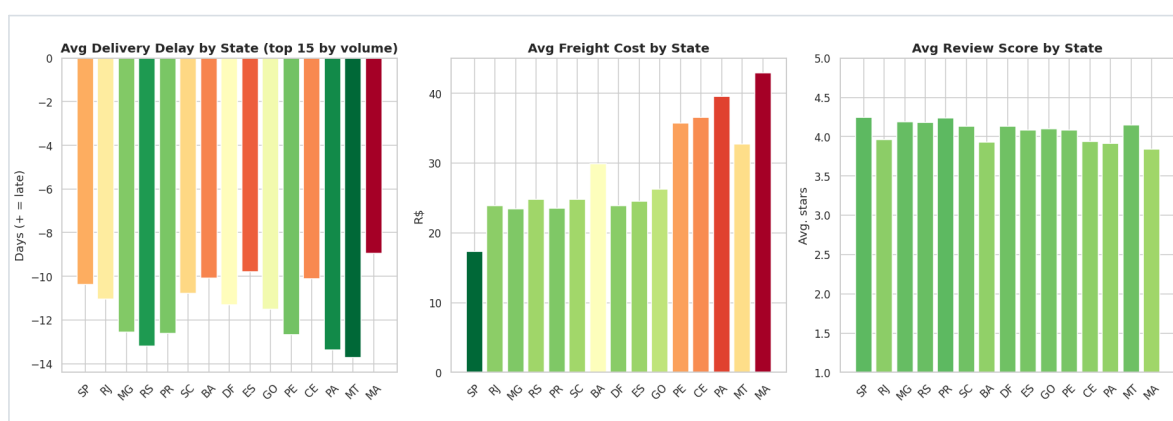


Fig. 13.2 — Average delivery delay, average freight cost, and average review score by state, top 15 by volume.



Fig. 13.3 — Freight cost vs. customer–seller distance proxy (degrees lat/lng).

What the output is expected to reveal: States farther from the main seller hub (Southeast, especially São Paulo) are expected to show longer delays, higher freight, and lower reviews simultaneously — a compounding disadvantage suggesting that regional seller recruitment or logistics hubs would directly improve the underserved regions' experience.

14 Payment Behavior Analysis

Guiding question: How do payment type and installment choices relate to order value, and do they appear connected to customer experience?

Analytical approach

- Compute payment-type mix (credit card / boleto / debit card / voucher) as a share of orders, and average order value by payment type.
- Plot average order value against number of installments chosen (credit-card orders) to test whether installment count is a basket-size proxy.
- Compare average review score across payment types to test whether payment method is a meaningful satisfaction lever.

Supporting visuals

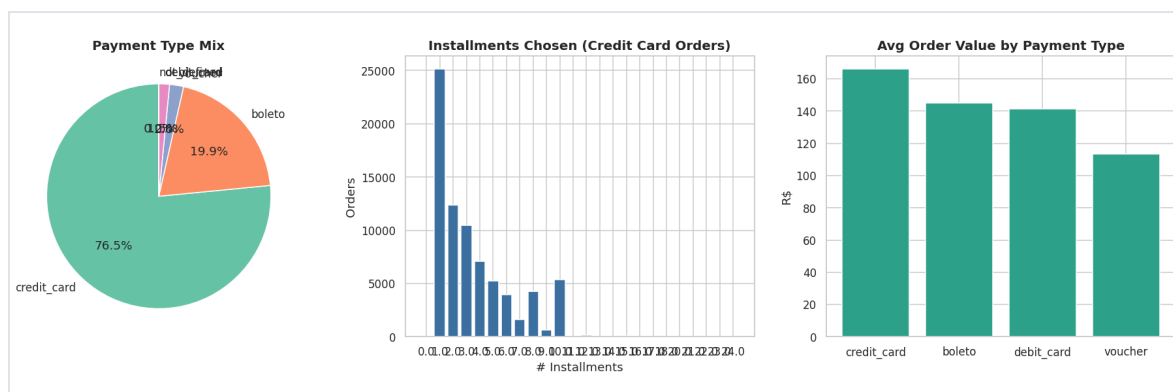


Fig. 14.1 — Payment type mix (left), installments chosen on credit-card orders (middle), and average order value by payment type (right).

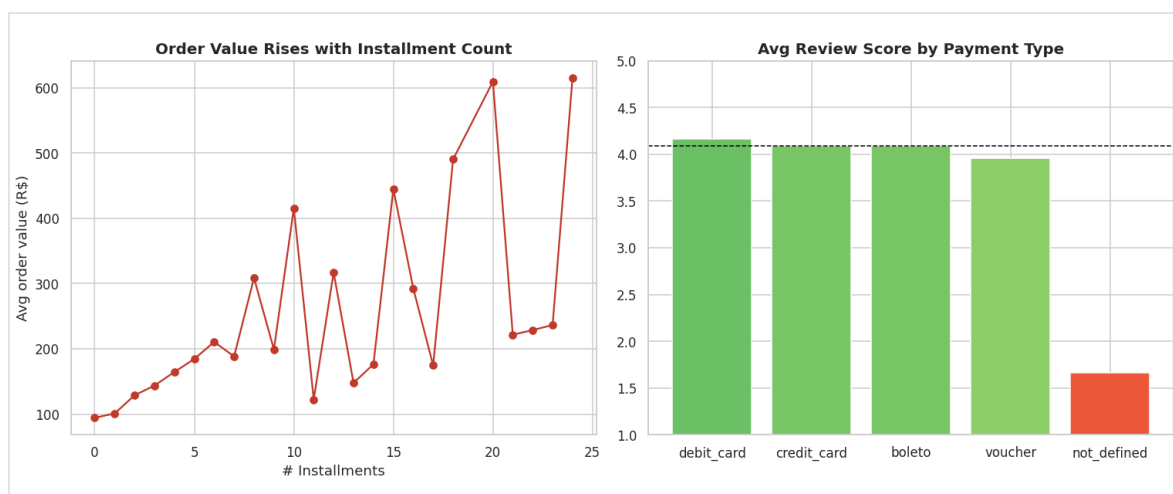


Fig. 14.2 — Average order value rises with installment count (left); average review score by payment type (right).

What the output is expected to reveal: Higher installment counts are expected to track with larger order values (a basket-size signal, not an independent behavior), while review-score differences across payment types should be modest — indicating payment experience is a secondary satisfaction factor, more useful to finance/merchandising than to CX teams.

15 Stage 4 — Statistical Analysis & Root-Cause Modeling

This closing stage synthesizes every dimension above into a defensible answer to Core Question 6: what most strongly associates with low review scores, and which factors are primary vs. secondary?

15.1 Correlation & Feature-Importance View

- Compute Pearson correlation of every engineered numeric feature (delivery delay, lead time, seller handling time, freight, price, installments, distance proxy, item count) against review score.
- Train a lightweight supervised model (e.g., Random Forest) purely to rank feature importance against a low-review target — not yet the production/business-facing model.
- Cross-reference the correlation ranking and the importance ranking; treat features that agree across both methods as more credible primary drivers.

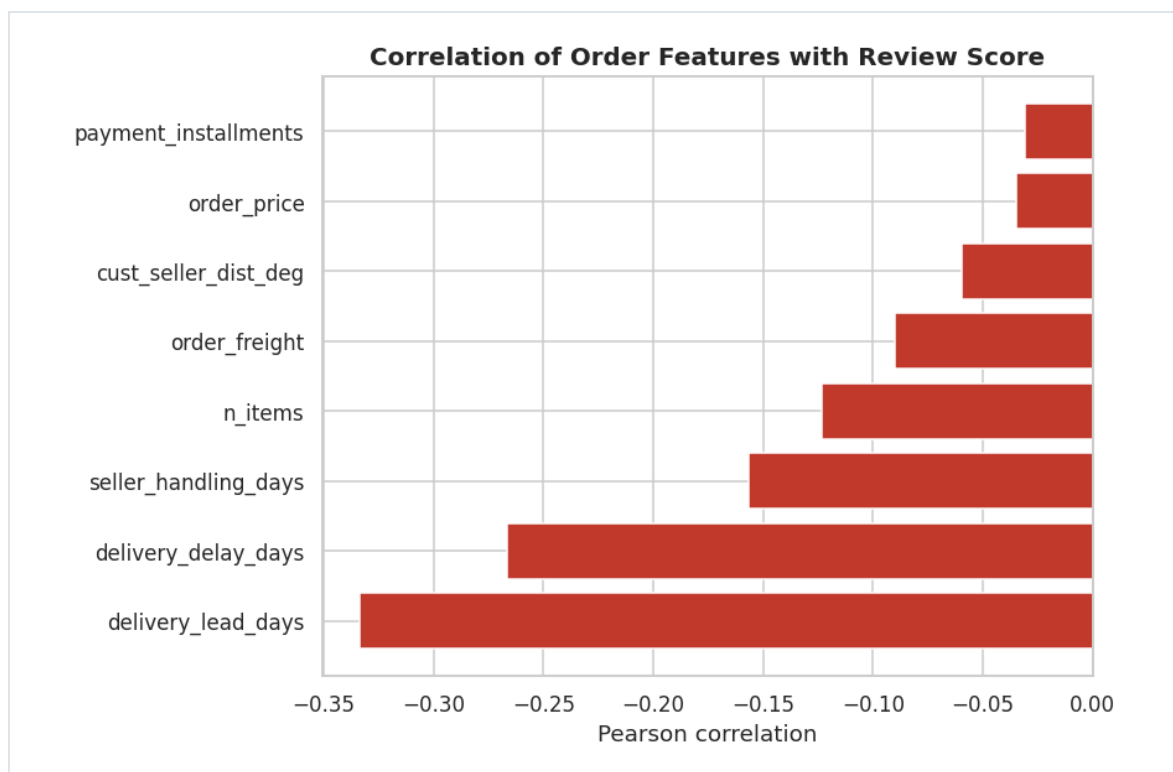


Fig. 15.1 — Pearson correlation of engineered order features with review score.

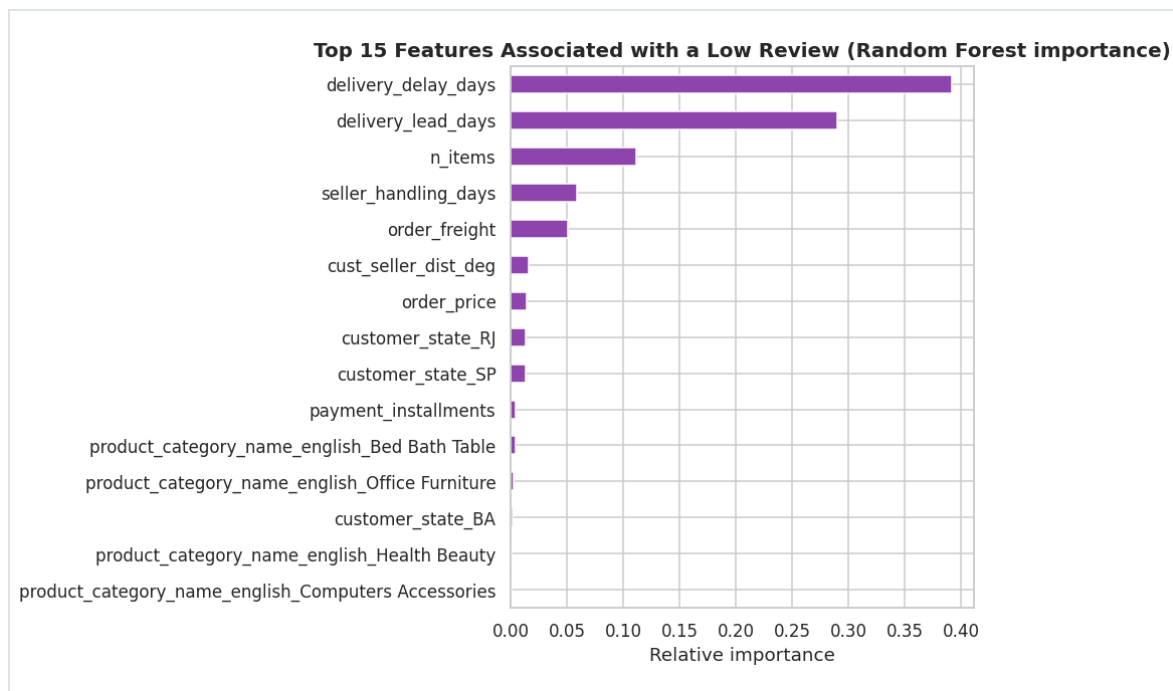


Fig. 15.2 — Top 15 features associated with a low review score (Random Forest importance).

15.1.1 Outlier & Anomaly Screening

A small tail of orders sits far outside normal delivery-time and pricing ranges. These are screened separately rather than silently clipped, since extreme-delay orders are disproportionately associated with 1-star reviews and may indicate genuine logistics failures (lost packages, carrier exceptions) rather than “normal” slow delivery; sellers evaluated on very small order counts can likewise show misleadingly perfect or poor averages, reinforcing the minimum-volume threshold used in the seller scorecard (Section 12).

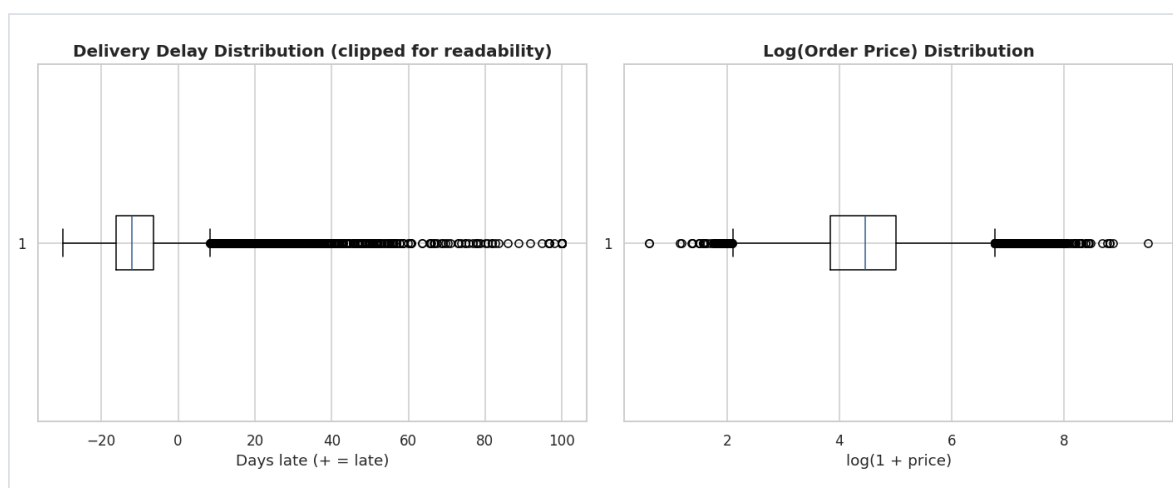


Fig. 15.3 — Delivery delay distribution and log(order price) distribution (clipped for readability), showing a long tail of outliers on both dimensions.

15.2 Predictive Model — “At-Risk Order” Early-Warning System

Because the satisfaction survey is sent only after delivery (or after the estimate passes), there is a window — while the order is still in transit, or just after carrier hand-off — in which Customer Experience could still

intervene. A classifier is proposed that flags at-risk orders using only information available at shipping time, deliberately excluding the actual delivery date/delay (which is only known after the fact and would leak the outcome).

- Features available at shipping time: seller handling time, order price/freight, item count, category, customer/seller state, distance proxy, payment type/installments.
- Evaluate with ROC/AUC and Precision-Recall curves (base rate of low reviews is a minority class, so PR curve matters as much as ROC), and a confusion matrix at a chosen operating threshold.
- Convert predicted probability into three operational risk tiers — High (~top 10%), Medium (~next 20%), Low (remainder) — so Customer Experience agents can prioritize outreach.
- Attach an illustrative, assumption-driven ROI simulation (clearly labeled as assumption-driven, not measured) to show leadership how to reason about the economics of proactive outreach.

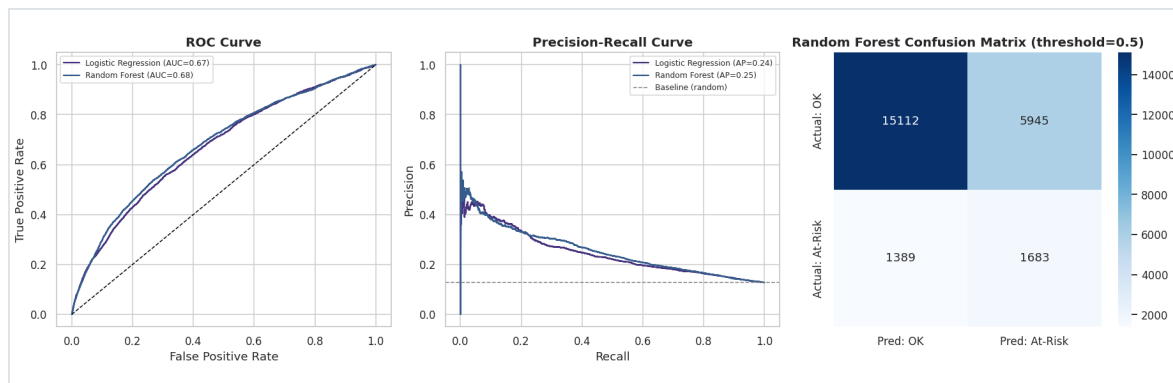


Fig. 15.4 — Model evaluation: ROC curve, precision-recall curve, and Random Forest confusion matrix at threshold 0.5.

Recommended validation path: (1) validate on a larger, real holdout period; (2) pilot the proactive-outreach workflow on the High-Risk tier only, with a control group, to measure true save rate rather than assume it; (3) feed seller-level risk-tier rates back into the seller scorecard (Section 12) so chronically high-risk sellers are addressed at the source.